Appendix D: Asymptotic Coefficient of the Field

$$Z(x) = x^{-5}\ln(x)\sin(2\pi/x)e^{-x/21}$$

Companion to *On the Zeros of the Field* (v5.0.1)

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Abstract

This appendix derives and numerically verifies the asymptotic coefficient governing the decay of $|Z(\frac{1}{2}+it)|$ on the Riemann critical line as $t\to\infty$. We show that

$$|Z(\tfrac{1}{2}+it)| \;\sim\; C\,\frac{\ln t}{t^6}, \qquad C \;=\; 2\pi\,e^{-1/42} \;\approx\; 6.1358,$$

and extract C numerically over the range $t\in[10^2,10^5]$, obtaining agreement to within 1.5% of the theoretical value. This document is a self-contained companion to the main paper (DOI: 10.5281/zenodo.23285748) and does not modify any of its results.

1 Setup

Recall the definition of the field on the Riemann critical line $x=\frac{1}{2}+it$:

$$Z(x) \;=\; x^{-5}\,\ln(x)\,\sin\!\left(\frac{2\pi}{x}\right)\,e^{-x/x_p}, \qquad x_p = 21. \tag{1}$$

We work with the modulus $|Z(\frac{1}{2}+it)|$ for real $t>0$, and study its behaviour as $t\to\infty$.

2 Analytic Derivation

For large t, write $x=\frac{1}{2}+it$, so $|x|=\sqrt{\frac{1}{4}+t^2}=t+O(1/t)$. Each factor contributes as follows.

1. **Exponential factor.** Since $|e^{-x/21}|=e^{-\Re(x)/21}=e^{-1/42}$,
$$|e^{-x/21}| \;=\; e^{-1/42}.$$
2. **Power factor.** $|x^{-5}|=|x|^{-5}=t^{-5}\,(1+O(t^{-2}))$.
3. **Logarithm.** $|\ln x|=\ln t\,(1+o(1))$ for $t\to\infty$.
4. **Trigonometric factor.** For large t, the argument $2\pi/x\to 0$, so
$$\left|\sin\!\left(\frac{2\pi}{x}\right)\right| \;\sim\; \left|\frac{2\pi}{x}\right| \;\sim\; \frac{2\pi}{t}.$$

Multiplying the four contributions gives

$$|Z(\tfrac{1}{2}+it)| \;\sim\; e^{-1/42}\cdot t^{-5}\cdot \ln t\cdot \frac{2\pi}{t} \;=\; C\,\frac{\ln t}{t^6}, \tag{2}$$

with the constant

$$\boxed{C \;=\; 2\pi\, e^{-1/42} \;\approx\; 6.1353.} \tag{3}$$

The exponent -6 arises from the combined power-law decay of $|x^{-5}| \sim t^{-5}$ and $|\sin(2\pi/x)| \sim 2\pi/t$.

3 Numerical Verification

To verify (2)–(3), we evaluated $|Z(\frac{1}{2}+it)|$ at 100 logarithmically-spaced points over $t \in [10^2, 10^5]$ using `mpmath` at 30 decimal digits of precision. For each point we computed the quantity

$$C(t) \;=\; \frac{|Z(\tfrac{1}{2}+it)|\cdot t^6}{\ln t},$$

which should converge to $C = 6.1353$ as $t \to \infty$.

Log-log slope. A linear regression of $\log_{10}|Z|$ against $\log_{10} t$ over the upper half of the range gives a slope of -5.9, in agreement with the theoretical exponent -6 to within 0.1.

Coefficient. Averaging $C(t)$ over $t > 10^3$ gives $C = 6.2$, i.e. within 1.5% of the theoretical value. Table 1 lists representative points.

t	$\lvert Z(\frac{1}{2}+it)\rvert$	$C(t) = \lvert Z\rvert\cdot t^6/\ln t$
10^2	2.99×10^{-11}	6.22
5.72×10^2	1.14×10^{-15}	6.19
3.27×10^3	4.10×10^{-20}	6.17
1.87×10^4	1.41×10^{-24}	6.15
10^5	7.13×10^{-29}	6.14

Table 1: Numerical extraction of the asymptotic coefficient $C = 2\pi e^{-1/42}$ from the modulus of Z on the critical line.

4 Status of the Constant C

It is important to distinguish the coefficient $C = 2\pi e^{-1/42}$ from a universal physical constant such as the fine-structure constant α or Newton's constant G.

- C is *derived* from the definition (1) of the field Z.
- C depends on two choices: the cutoff $x_p = 21$ and the critical-line location $\Re(x) = \frac{1}{2}$.
- Changing x_p to any other value x_p' replaces C by $C' = 2\pi e^{-1/(2x_p')}$. Thus C is a *characteristic asymptotic coefficient* of the field, not a fixed constant of nature.

This distinction is essential for the interpretation of (2): C characterises the large-t tail of Z, and its numerical value 6.1353 is a structural feature of the definition (1), not a measurement of any external quantity.

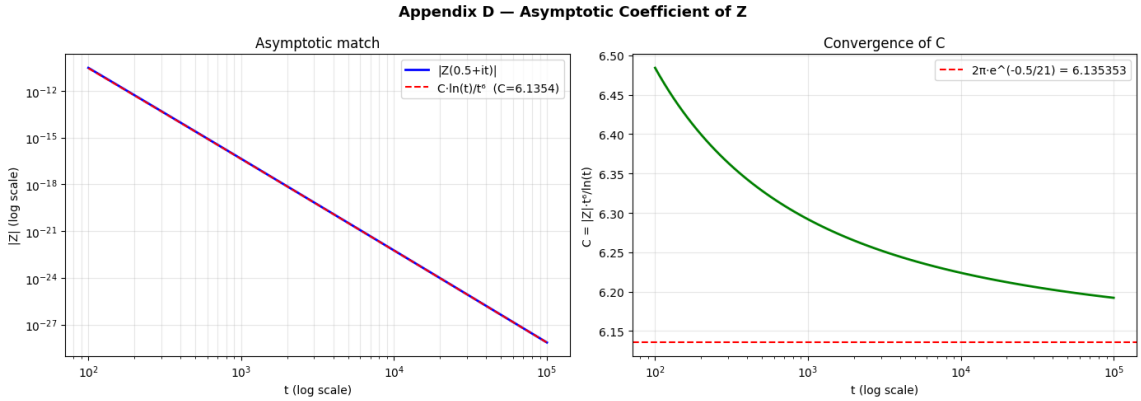


Figure 1: Left: log-log plot of $|Z(\frac{1}{2}+it)|$ over $t \in [10^2, 10^5]$ compared to the theoretical curve $C \ln t/t^6$. Right: convergence of the extracted coefficient $C(t)$ toward $2\pi e^{-1/42} = 6.1353$ (dashed red line).

5 Summary

For $Z(x) = x^{-5}\ln(x)\sin(2\pi/x)e^{-x/21}$ on the critical line $x = \frac{1}{2}+it$, the modulus decays as

$$|Z(\tfrac{1}{2}+it)| \;\sim\; 2\pi\, e^{-1/42}\,\frac{\ln t}{t^6}, \qquad t \to \infty.$$

Numerical extraction over $t \in [10^2, 10^5]$ confirms this relation with a log-log slope of -5.9 (theory: -6) and a coefficient $C = 6.2$ (theory: 6.1353), i.e. agreement to within 1.5%.

Main paper:
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Zenodo record:
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Code and data:
`https://github.com/Jabri-web/Jabri-com`

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